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Displacement Due to Violence in Colombia: A Household-Level Analysis

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I. Introduction

The past decade has shown an increased occurrence of civil wars and ethnic conflicts (Kaldor 1999). As a consequence, millions of people have left their homes, either as refugees seeking asylum abroad or as internally displaced persons (IDPs) who seek shelter within their home country's borders.¹ The U.S. Committee for Refugees (USCR) estimates that, as of the end of 2002, there were 13 million refugees and 21.8 million IDPs in the world (USCR 2002).² While legal instruments have been established that empower the United Nations and the international community to provide protection and support to refugees, the situation of IDPs is particularly complicated by the fact that the national government remains the primary institution in charge of assistance to IDPs.³

This study was initiated while we were employed at the Faculty of Economics of the University of the Andes in Bogotá, Colombia. The research project was funded by Colciencias and the National Department of Planning, Colombia. We would like to express our immense gratitude to our colleagues in the project, Alberto Galán and Jaime Andrés Erazo, in Colombia. We also thank Oded Stark for a very interesting discussion on the issue of displacement, which has enriched our theoretical considerations and ideas for future research. Last, but not least, we thank Weselina Angelow, Max Holtmann, Kerstin Becker, and Quixia Zhu for excellent research assistance and Christina Endemann for help with the final editing.

¹ The United Nation's Guiding Principles on Internal Displacement (UNCHR 1998) define internally displaced persons as "persons or groups of persons who have been forced or obliged to flee or to leave their homes or places of habitual residence, in particular as a result of or in order to avoid the effects of armed conflict, situations of generalized violence, violations of human rights or natural or human-made disasters, and who have not crossed an internationally recognized State border."

² Statistics on forced migration are difficult to obtain and therefore are often inexact. The Norwegian Refugee Council estimates the number of internally displaced persons (IDPs) in the world at 25 million (Norwegian Refugee Council 2003).

³ According to the Norwegian Refugee Council (2003), Africa has the highest number of IDPs (13.5

International human rights law considers forced displacement to be a human rights violation. Moreover, the consequences of the phenomenon for the affected countries are severe. Increases in unemployment, the violent redistribution of land ownership, strong inefficiencies in resource allocation, and the effects of large and unplanned demographic inflows into cities that act as receptors of the displaced are just a few of the many problems caused by displacement. Interest in attempting to address the needs of IDPs has grown steadily over the past few years among a number of United Nations, international, and nongovernmental organizations, as well as national governments, which have responded by developing assistance programs (UNDP 2003).

The vast majority of these programs have focused on the consequences of displacement, that is, they have attempted to help the displaced to deal with the situation of arriving in an unknown place. However, the continuing and increasing trend of displacement and the limited ability of the receptor cities to absorb these immense masses of people are making it more and more urgent to design policies of prevention, assistance, and resettlement. A *sine qua non* condition for defining such policies is to understand the decision-making process underlying displacement. By definition, violence is the trigger of the kind of displacement considered here. However, people react differently to given levels of direct and indirect violence. Frequently we observe that a substantial portion of the population in areas of violence decides to stay despite the risk that this implies for them. Although some hypotheses about the determinants of displacement decisions have been put forth in the literature, the relative importance of these factors and their direction and strength at the household level have not been analyzed in a rigorous manner.⁴ Is displacement caused solely by violence or do sociodemographic characteristics also play a role? Is violence indiscriminate or is it targeted to certain types of households? Can some policies, in addition to the end of the conflict, prevent displacement?

To our knowledge, this article presents the first attempt to identify the determinants of displacement at the household level using rigorous statistical analysis. Specifically, we aim to provide answers to the above questions, focusing on the case of Colombia and utilizing household-level surveys to estimate a random utility model of displacement. According to USCR statistics, Colombia continuously ranks among the 10 countries with the highest incidence of

million), followed by Asia and the Pacific (4.6 million), the Americas (3 million), Europe (3.2 million), and the Middle East (1.5 million).

⁴ For previous work on displacement see, e.g., Deng (1995). Analyses of displacement in Colombia have been conducted by Consultoría para los Derechos Humanos y el Desplazamiento (CODHES 1996, 1997, 1998), Conferencia Episcopal de Colombia (1995, 1999a, 1999b), and several other qualitative studies. For a detailed review of this literature, see Erazo et al. (2000).

displacement worldwide (USCR 2002).⁵ The armed conflict over economic and political power fought between the country's guerrilla groups, paramilitary vigilantes, and armed forces has forced at least 1,800,000 Colombians, or 4.3% of the population, to leave their homes (Arboleda and Correa 2002).⁶ During the new millennium, displacement has intensified, increasing 38% between the years 2000 and 2001. Almost 74% of Colombian municipalities have experienced expulsion or reception of displaced population (Red de Solidaridad Social 2002). The outlook for the future is dim. Peace talks recently came to a halt, the government elected for the period 2002–6 (and recently reelected) aims to defeat rebel groups using military confrontations, and human rights violations escalated during the years 2002 and 2003.

The structure of this article is as follows. Section II discusses similarities and differences between existing theories of migration and the context of displacement. It then proceeds to develop a framework for the empirical analysis of the displacement decision. Section III discusses the data used for our empirical analysis and the specific hypotheses developed in the context of this data. Section IV presents an econometric analysis of the displacement decision. Finally, Section V concludes and discusses the policy implications of our analysis.

II. A Framework for Analysis of the Displacement Decision

Our objective is to analyze the determinants of displacement at the household level. Probably because displacement is perceived as a political rather than an economic problem (Lucas 1997), the contributions of the economics literature to analyze it are scant, and data availability is scarce. A few existing studies include the occurrence of violent conflict and natural disasters in analyses of aggregate migration patterns. In an econometric analysis of internal migration in Colombia during the period spanning from 1951 to 1964, Schultz (1971) finds that net migration rates are increasing in the incidence of politically motivated homicides. Morrison and May (1994) show that violence was a key determinant of migration in Guatemala during the most violent period of the civil war. An analysis of migration from the north of China to Manchuria by Gottschang (1987) indicates that disasters, such as wars, famines, and floods,

⁵ According to USCR's *World Refugee Report, 2002*, in 2002 Colombia ranked sixth among what they call the "Top 10 Refugee Hot Spots."

⁶ The USCR's estimates are even higher, estimating the total number of uprooted people (including refugees and IDPs) at 2.5 million, with more than 400,000 of these newly displaced in 2002 (USCR 2002). The nongovernmental organization (NGO) CODHES estimates that 2.8 million have been displaced since 1985 (*Economist*, April 19, 2003, 46).

among many others, promote migration when they occur but do not explain the persistent migration between the two regions.

However, these studies do not focus systematically on the displacement decision at the household level. While the flight of refugees and displaced persons can be seen as extreme cases of migration settings, there are important structural differences between the displacement decision and a household's decision to migrate in the absence of violence. Given the void in the literature, the purpose of this article is to develop and apply a framework for empirically analyzing the determinants of the displacement decision at the household level. As stated earlier, we argue that the displacement decision cannot be explained by the presence of violent activities alone. Although violence in one form or another triggers displacement, it does not seem to be the only factor in determining whether or not a person leaves her place of origin. In particular, we often observe that a substantial proportion of the population in areas with high occurrence of violence stays there despite the risk that this implies.

In general, we can think of two reasons for this phenomenon. First, those households that decide to stay may confront a lower risk of directly suffering from the violence (including death) than those that decide to leave. If so, we would like to know what determines this (perceived) risk. Second, other factors might influence the displacement decision. At the moment of making the decision whether to leave their homes or remain at their place of origin, households have to consider the potential risks and benefits of the two options and choose the one that is best for them—or, perhaps more appropriately, choose the lesser of two evils. In doing so, individuals have to form expectations about their quality of life at both the place of origin and the potential destination in case of displacement. In this sense, the displacement decision resembles other migration decisions. However, the important role of violence in displacement decisions introduces significant differences that need to be addressed.

In what follows we discuss the main results of existing theories of migration (not related to violence) and their applicability to the displacement context. In particular, we highlight useful similarities but also important differences to be considered in a model of displacement decisions. We then present a framework for the empirical analysis of the displacement decision, which we will proceed to apply in Section III of this article.

A. The Theory of Migration and the Displacement Decision: Differences and Similarities

We start by briefly reviewing the existing theories on migration decisions in order to highlight important similarities and differences to the displacement decision. Early theories of migration emphasized differences in salary levels as

the principal cause of migration (e.g., Ravenstein 1885, 1889; Lewis 1954; Ranis and Fei 1961; Todaro 1969). Later, human capital theory stated that migration decisions are dependent as well upon individual characteristics,⁷ such as the individual's planning horizon and thus his age (Plane 1993), personal contacts at the receptor location and access to information (Yap 1977), the level of education (Greenwood 1997), and the individual's level of risk aversion. Imperfect and costly information may also influence the migration decision (e.g., Stark and Levhari 1982; Maier 1985; Dustmann 1997). While basic information costs may deter migration, information about economic and social opportunities at the receptor location may encourage migration (Fischer, Martin, and Straubhaar 1997).

The above concepts from migration theory are also relevant to the displacement decision. However, additional factors influence the latter, and, for some traditional determinants of migration, the effect may change when considering displacement. For example, displacement may very well take place in spite of negative income differentials or a decrease in household wealth and living conditions after displacement if, instead, the risk of suffering from a violent act diminishes with displacement. Contrary to the case of migration for purely economic reasons, individuals with lower risk aversion in a context of violence could be more disposed to stay at their place of origin and to incur risks of suffering from violent action. Moreover, better access to information might decrease the propensity for displacement because knowing the precarious situation of already displaced people may dissuade some people from leaving their homes.

The new economics of migration moderates the assumption of individuals being autonomous without any social context by considering individuals as an integral part of the household (e.g., Stark and Bloom 1985; Stark 1991). This helps to explain why we often observe some members of the household migrating while others stay at the place of origin, an issue that is also relevant to the displacement problem. In some cases, the household head leaves the place of origin first until minimum conditions at the receptor location are achieved before the entire family follows. In other cases, young household members migrate first to reduce the risk of forced recruitment by rebel groups. The household is then confronted not only with the decision to displace but also with the decision as to which members of the household should leave the place of origin. Unfortunately, the data available for this study do not permit such an intrahousehold analysis.

⁷ Some examples are Sjaastad (1962), Becker (1964, 1975), Mincer (1974), Todaro (1976, 1989), and Todaro and Maruszko (1987).

In an interdisciplinary study, Fischer et al. (1997) present determinants that contribute to limit migration. These include the loss of specific assets at the place of origin when migrating (Lucas 1997), local specific knowledge (Jaganathan 1987), social assets, and potential discrimination faced at the receptor location.

These considerations are also relevant to the decision-making process regarding forced displacement. Discrimination and marginalization are common phenomena encountered by the displaced in receptor locations. The loss of location-specific assets considered by the household may increase or decrease as the result of violent conflict. On the one hand, the prevalence of conflict at the place of origin can be expected to lower the market demand for household assets, leading to difficulties in selling these assets. This would raise the household's opportunity cost from migrating as compared to the situation without violence.⁸ This particularly would be the case if the probability of seizure of household assets by armed groups were to increase when these assets were abandoned. On the other hand, violent actors may force households to abandon their properties anyway, regardless of their decision to leave the area. In this case, the effective loss of assets implied by displacement falls to zero. An important difference between the contexts of displacement and conventional migration, however, is that assets and roots may affect not only the costs of displacement but also the likelihood of becoming a victim of direct threats. This aspect will be further discussed below.

Finally, Fischer et al. (1997) note that the concept of bounded rationality may be useful in understanding migration. Specifically, Simon (1957, 1983) postulates that human actions are limited by the situations individuals are confronted with and with their previous experiences, emotional patterns, and limited abilities to calculate the costs and benefits of a decision. The effects of bounded rationality are similar to those of incomplete information. An additional implication is that individuals face difficulties using all the information available when the information relevant to a particular decision widely differs from the information an individual usually processes. These ideas are particularly applicable to the displacement issue. The population living in zones of conflict is subjected to tremendous stress imposed by violence and personal threats. As a consequence, we would not expect them to take completely rational decisions and consciously evaluate all costs and benefits related to displacement, particularly when faced with immediate personal threats. One

⁸ The opportunity cost of displacement is given by the value of the household's assets in the place of origin if the household stays in the area minus the value of those same assets if the household opts for displacement.

consequence is that violence may have an impact on the effect of socioeconomic factors on household decisions.

B. A Basic Framework for the Analysis of the Displacement Decision

We now proceed to develop a framework for the empirical analysis of displacement decisions. Our previous discussion has shown that, although migration theory constitutes a useful starting point, there exist important differences between migration caused by economic reasons and displacement caused by violent action. First, violence introduces additional elements of risk, including the danger of losing one's life, the death of other household members, the possibility of suffering some other kind of violent act, and the risk of losing property. Although economic reasons may play a secondary role, these risks are likely to represent the principal motivations for displacement. Second, individuals facing high levels of violence are subject to tremendous levels of stress. Hence, decisions may be taken impulsively and without adequate information. As noted above, these differences introduce additional factors to be considered in the empirical analysis and also may change some of the hypotheses about the direction of the effects of the factors considered in migration analysis. Ignoring these important differences between the displacement decision and standard migration theory could seriously bias our understanding of migration decisions taken in the context of violence.

To address these differences, we propose a model where the decision to migrate depends on safety perceptions in addition to the variable categories highlighted by the standard migration literature. Moreover, the specific determining factors in each category are adjusted to the displacement context, and hypotheses on the expected direction of effects are adapted or changed accordingly. In addition, the model allows for the idea that the impact of economic factors on the migration decision varies with the degree of violence faced by a household. Finally, the model considers the fact that households' safety perceptions depend on direct and indirect risks of suffering from violence and that some of these risks may be endogenously dependent on households' social and economic characteristics.

Modeling the Displacement Decision

To start, consider the following very general way of presenting the displacement decision. Household i decides to displace if the expected utility from displacement is greater than the expected utility of staying in the place of origin, or

$$EU_{id} > EU_{in}, \quad (1)$$

where U_{ij} is the indirect utility function of household i at place j , with $j = d$

denoting the place of reception and $j = n$ denoting the place of origin, and E is the expectations operator. We can rewrite the expected utility as

$$EU_{ij} = v_{ij} + \varepsilon_{ij}, \quad (2)$$

where v_{ij} is the observable utility and ε_{ij} is a random term with a mean of zero. The random term includes all unknown or not quantifiable variables.

The observable utility of staying at the place of origin or displacing to the place of reception depends on a variety of factors. First of all, the perception of the safety level at the places of origin and reception is an important determinant of the utility function. Second, the income and quality of life at the two locations affect the level of utility. Third, migration and information costs, such as loss of goods at the place of origin or family ties at the receptor location, are important determinants for the estimation of the costs and benefits of displacement. Finally, the migration propensity might be affected by sociodemographic characteristics of the household, as these determine the household's preference structure. The latter will determine how the household evaluates the trade-offs between the increased security from violence and the riskiness of living conditions after displacement.

Thus, we write the observable utility as

$$v_{ij} = f(S_{ij}, Y_{ij}, C_i, Z_i), \quad (3)$$

where S_{ij} is the perception of safety for household i at place j , Y_{ij} is the economic status for household i at place j , C_i stands for migration and information costs, and Z_i represents household characteristics that influence preferences.

From equations (1), (2), and (3), the probability of displacement can be written as

$$\Pr_i(D_i = 1) = g(S_{id}, S_{in}, Y_{id}, Y_{in}, C_i, Z_i), \quad (4)$$

where $D_i = 1$ indicates that household i decides to migrate, and $D_i = 0$ indicates that the household stays at its site of origin.

Perceptions of safety generally depend on the overall level of violence in the area of origin (indirect threats, T_i^I) as well as the degree to which households feel directly threatened by this violence (T_i^D):

$$S_{in} = h(T_i^I, T_i^D). \quad (5)$$

As stated earlier, one possible reason why we observe some households staying in their area of origin despite high incidences of violence is that they feel less threatened personally by the violence than those who decide to leave. One important indicator of T_i^D is whether any of the household members have received direct death threats or other threats. However, we would expect that

the occurrence of direct threats to the household is itself endogenous and depends on household social and economic characteristics. For example, more wealthy households or those playing an important role in the community may be more likely to be the target of direct threats. This is because the armed groups themselves may follow specific strategies. The economic gains to the violent actors from the displacement of wealthier households and subsequent expropriation of their land and other assets increase with the wealth level of the household. Moreover, by threatening households that play a more important role in the community, armed groups can set an example for the rest of the population and thereby save on subsequent acts of intimidation to others.⁹ Finally, the likelihood of threats depends on the presence of armed groups in the region. This includes subversive groups as well as government forces, the latter of which may provide enhanced security but may also contribute to escalations of violence in the area.

Thus, we stipulate that the probability of a household being threatened directly can be written as

$$T_i^D = l(E_i, V_i, A_i) + e_i, \quad (6)$$

where E_i denotes the economic status of household i , V_i stands for ties of household i with the place of origin/importance of household i in the community, A_i represents the existence of armed actors in the region of origin of the household, and e_i is a random term.

Note that, as explained below in the discussion of specific proxies, all explanatory variables in equation (6) are also likely to affect the displacement decision in equation (4) as they are subcomponents of the independent variables in (4). While conceptually the predicted probability of direct threats should be used as a variable in the displacement regression, we thus lack an exclusion restriction that would allow us to estimate this structural form of the model. We therefore estimate the reduced-form equation for the probability of displacement:

$$\Pr_i(D_i = 1) = \tilde{g}(T_i^D, S_{id}, Y_{id}, Y_{in}, C_i, Z_i). \quad (7)$$

To shed further light on the structural relationships, we also report below the econometric analysis of the determinants of being threatened. Comparing both regressions permits us to draw some inferences about the direct effects of some variables on the displacement decision (e.g., the effect of household

⁹ A very interesting venue for future work would be to model this issue formally as a two-player game between the armed groups and the population of the community.

economic status through living standards) and the indirect effect of these variables (i.e., the effect of economic status on perceptions of security).

Two Types of Displacement and the Potential Importance of Interaction Terms

We would expect that safety perceptions not only influence the expected utilities directly but also may affect the degree to which other variables, particularly economic incentives, are considered in the decision to migrate. Specifically, we can distinguish between two types of displacement, **preventive and reactive**. Reactive displacement refers to those people faced with high levels of violence who tend to leave the region impulsively because they see no other option to save their life. Preventive displacement refers to those households that, given a lower presence of violence in the region, consider the risks involved in staying or leaving and then decide for either of these options.

Obviously, the factors other than violence itself would be expected to play a more important role for the latter group than for the former.¹⁰ One option for testing this hypothesis is to use interaction terms between the variable reflecting the level of violence and at least some of the other (particularly the economic) variables. If our hypothesis holds, the interaction terms should be significant and reduce the effect of economic variables on the displacement decision. The results presented below also shed some light on the impact of violence on the general urbanization trend observed in Colombia and other countries.

III. Data, Hypotheses, and Descriptive Analysis

A. The Data

In order to analyze the determinants of displacement, we conducted a household survey during the year 2000 of a sample of people who decided to leave their area of origin due to reasons of violence (hereafter referred to as “displaced”) and those who remained in areas of violence (hereafter referred to as “nondisplaced”). We administered 376 household surveys. Specifically, a total of 200 household interviews were conducted in three major receptor cities: Bogotá, Medellín, and Cartagena.¹¹ The sample was focused on identifying displaced people from the departments of Antioquia and Córdoba, as these are two of

¹⁰ There are several reasons for this. First, people faced with high levels of violence may be acting boundedly rational due to the high amount of stress imposed by violence. Second, violence may modify the returns to some assets.

¹¹ The distribution of the surveys was as follows: 100 in Bogotá, 50 in Cartagena, and 50 in Medellín.

TABLE 1
WEIGHTED AVERAGES OF BASIC ECONOMIC AND VIOLENCE INDICATORS FOR MUNICIPALITIES OF ORIGIN OF
DISPLACED AND NONDISPLACED HOUSEHOLDS SAMPLED

	Expulsion Municipalities of Displaced Households Interviewed	Municipalities Where Nondisplaced Households Were Interviewed	t-Statistic
Percentage of households in municipality with un- met basic needs in 2000	61.14	65.32	-1.6049
Percentage of rural territory (1993)	58.71	59.12	-.3655
Homicide rate in 2000 (number of deaths per 100,000 inhabitants)	64.72	80.00	-1.2642
Number of kidnappings per capita 1993-99	6.08	18.63	.7126

Source. Own calculations based on municipality-level data collected at Centro de Estudios sobre Desarrollo Económico (2005).

Note. Weights were calculated according to the percentage of households in the sample in each municipality.

the principal zones of expulsion.¹² For the surveys of the nondisplaced, three communities in these two departments that showed a constantly high rate of displacement were selected: Apartadó and Mutatá for the department of Antioquia and Tierralta for Córdoba. A total of 176 surveys of households remaining in these communities were conducted.¹³ The three municipalities selected for the nondisplaced survey appear as expulsion municipalities for 22.3% of the displaced households in the sample. To test whether the samples of displaced and nondisplaced households are comparable or whether they are systematically different, we compare the weighted averages of basic economic and violence indicators for expulsion municipalities of the displaced sample with those for the municipalities in the nondisplaced sample.¹⁴ The results, presented in table 1, indicate that the municipalities of origin in the two samples appear to be reasonably comparable. In particular, the *t*-statistics show that differences are not statistically significant. The survey included a wide variety of questions on sociodemographic characteristics, living conditions,

¹² For 1998, the department of Antioquia was the origin of 20.65% of the displaced people nationwide, and 8.37% came from the department of Córdoba. For further details regarding the survey and sample, see Erazo et al. (2000). Nearly 90% of households from the displaced sample migrated from expulsion municipalities located in the two departments of nondisplaced communities, Antioquia and Córdoba. A municipality is the smallest administrative unit in Colombia.

¹³ The distribution of households surveyed in the three communities was as follows: 30 in Mutatá, 100 in Apartadó, and 46 in Tierralta. For further details regarding the selection of the samples, see Erazo et al. (2000).

¹⁴ Weights were calculated according to the percentage of households in the sample in each municipality.

TABLE 2
DESCRIPTIVE STATISTICS FOR DISPLACED AND NONDISPLACED HOUSEHOLDS

Variable and Definition	Displaced Population		Nondisplaced population	
	Mean	Standard Deviation	Mean	Standard Deviation
Direct threats (equals 1 if any household member received death threats, 0 otherwise)	.78	...	.09	...
Indirect violence (equals 1 if respondent knows of other incidences of violent acts in village of origin or nearby villages, 0 otherwise)	.98	...	.75	...
Military presence at place of origin (dummy variable)	.91	...	.90	...
Police presence at place of origin (dummy variable)	.58	...	.90	...
Paramilitary presence at place of origin (dummy variable)	.97	...	.66	...
Guerrilla presence at place of origin (dummy variable)	.96	...	.50	...
Land ownership at place of origin (dummy variable)	.80	...	.06	...
Standardized land size at place of origin	-.22	.12	.01	.02
Access to social services at place of origin (dummy variable)	.34	...	.91	...
Highest level of education in household (years)	7.37	.49	9.01	.30
Number of livestock owned at place of origin	.66	.36	.01	.01
Access to public utilities (dummy variable)	.36	...	.99	...
Access to media (number of categories of public media (radio, newspaper, television, periodicals, telephone, others) accessible to household at the place of origin)	2.79	.42	3.35	.11
Age of household head	33.71	1.99	43.93	1.11
Gender of household head (equals 1 if male, 0 if female)	.90	...	.59	...
Household membership in organizations at place of origin (number of organizations)	.26	.13	.27	.04
Contacts at reception site (dummy variable)	.87	...	.74	...
Years of residence at place of origin	15.53	4.10	21.01	1.05

Source. Authors' calculations based on survey data.

access to support, current needs, and more. Definitions of variables used in the econometric analysis and associated descriptive statistics are presented in table 2 and further discussed below. In each case, interviews were conducted with the household head. It should be noted that the sample is not representative of the Colombian displaced population; hence, results are not generalizable. To correct for the potential bias induced by our choice-based sampling approach, we use the weighted exogenous sampling maximum likelihood estimator suggested by Manski and Lerman (1977). Further details on this approach are presented in the appendix.

B. Variables and Hypotheses

The data from the survey described above are used to estimate the probability of displacement (eq. [7]) and the probability of being a victim of a direct threat (eq. [5]). We now describe the proxies used for the general categories of factors specified in the conceptual model and our corresponding hypotheses. Detailed variable definitions are provided in table 2. It should be noted up front that some variables (e.g., age and education) serve as proxies for several factors. If the hypothesized direction of their effect on the displacement decision differs, the aggregate impact of these variables may be ambiguous and the hypotheses cannot be tested in detail.

Determinants of Displacement

Among the variables listed in equation (7), perceptions of safety at the place of origin (S_m) are of particular relevance. In general, we expect that households perceiving greater levels of insecurity are more inclined to displace. Perceptions of safety, albeit not observable, can be approximated by a set of proxies:

Violent acts suffered by family members, friends, neighbors, or people of other neighborhoods. The occurrence of such acts increases the household's perceived insecurity, raising the probability of displacement. As discussed earlier, the variable representing direct threats cannot be included in the analysis as it is endogenous and dependent on the same set of variables used as independent variables in the displacement regression. We include a measure for indirect violence (a dummy equal to one if respondents knew of other incidences of violent acts in their own or nearby villages).

Presence of subversive groups. We hypothesize that perceived insecurity and thereby the propensity to opt for displacement increase where guerrilla and/or paramilitary groups are present in the area of origin.

Government presence in the region (military or other security forces). It is not clear a priori whether government presence increases or decreases the perception of security in violent areas. On the one hand, government presence may provide protection from subversive groups. On the other hand, it may increase the likelihood of the civil population being caught in the middle of the battle between the government and subversive groups. Moreover, incidents of government troops conducting human rights violations exist as well.

We would expect the probability of displacement to increase with economic status at the receptor location (Y_{id}) and decrease with economic status at the place of origin (Y_{in}). Economic status at the receptor location can be approximated with the following variables:¹⁵

¹⁵ Unfortunately, the data collected did not permit measuring incomes directly. The usual difficulties

Characteristics of the household head (age, education). In accordance with the migration literature, we would expect younger household heads to have a higher propensity for displacement because their planning horizon, and, therefore, the stream of discounted income in the place of reception is higher. We would also expect that a higher level of education is associated with a higher probability of displacement because it increases the expected income in the urban setting of the receptor location. However, as we will see below, education might also have impacts on information provision and preferences that may counteract this effect.

Personal contacts at the place of reception. We do not have an a priori expectation on this effect. On the one hand, in line with the migration literature, personal contacts at the place of reception may increase the expected standard of living there because it can provide a potential support network in finding employment and housing. On the other hand, personal contacts may lower unrealistically optimistic expectations regarding the employment possibilities, living standards, and access to government support in the receptor locations.

The proxies for Y_{in} , the economic status at the place of origin include:

Land possessed by the household at the place of origin. Land ownership is expected to deter displacement. Displacement impedes households with land to continue agricultural production; thus, income may decline substantially. The opportunity cost of idle land is greater for larger plots of land; therefore, the probability of displacement is expected to decrease with the amount of land possessed.

Access to educational services and health services at the place of origin. A better access to these services is associated with a higher standard of living and is therefore expected to reduce the probability of displacement.

Connection to public utilities. Households connected to public utilities have a higher standard of living at the place of origin and as such are expected to have lower probabilities of displacement.

Livestock ownership. Livestock can be both a direct source of income and a form of savings. In the sense that greater numbers of livestock are associated with greater economic status at the place of origin, greater livestock numbers are expected to decrease the probability of displacement. However, livestock is an asset that can be more easily sold than other assets such as land. In this sense, greater numbers of livestock may also represent greater available savings

of collecting income data are aggravated in the case of displacement by the high degree of informal sector activity, the high percentage of households with income from agricultural production, and the time lapses between displacement and the survey. Potentially, there may also have been strategic overstatement (understatement) of incomes before (after) displacement motivated by the hope that the survey results would affect actual compensations and other support programs.

to use during the displacement process and as a consequence may increase the probability of displacement.

Characteristics of the household head (age, education). The considerations here are exactly the counterparts of the ones discussed for the income at the receptor place.

Migration and information costs (C_i) are approximated with the following variables:

Amount of land possessed by the household at the place of origin. The possession of location-specific assets implies a potentially higher cost of displacement because the latter usually implies a loss of these assets. These higher migration costs are expected to reduce the propensity for displacement.

Linkage to the region of origin (years lived in the region). A stronger linkage to the place of origin implies a higher emotional cost of leaving this way of life behind, thereby deterring displacement.

Education of the household head. A higher education level may lower information costs, but again the impact on displacement is ambiguous, depending on the kind of information obtained.

Contacts at the place of reception. Similarly, contacts at the place of reception may improve information on actual living conditions there and potentially deter displacement. The counteracting effect on adjustment costs was already discussed above.

Access to the media. Once again, the effect will depend on the type of information provided and is not clear a priori.

Household characteristics (Z_i) are also included to reflect the household's preference structure (e.g., utility function, discount rate, and risk aversion). Age and sex of the household head are incorporated as household's characteristics. Younger household heads may be less risk averse than older ones, and female household heads may be more risk averse than male ones. However, contrary to the migration literature, risk aversion may not deter displacement but may rather spur it due to the high risks to the household's security in the region of origin. The impact of education on preference structure is less clear a priori.

Finally, to test the hypothesis that violence may reduce the role of economic incentives in the displacement decisions, we include interaction terms between the variable "indirect violence" and important economic indicators (land ownership, land size, access to social services, and education) in the regression.

Determinants of Direct Threats

As explained earlier, we also analyze the determinants of direct threats to shed some light on the structural relationships specified in Section II.B. For com-

pleteness, we include the same variables used in the displacement regression in the econometric analysis of determinants of direct threats. The dependent variable is a dummy variable equal to one if any household member has received death threats and zero otherwise. From our discussion in Section II.B, we expect the following variables to be particularly important in determining direct threats:

Households with greater economic status (E_i) are more likely to be threatened by armed groups. Thus, from this perspective, wealthier households are hypothesized to be more likely to opt for displacement. Note that this counteracts the direct effect of household wealth explained above, that is, wealthier households have more to lose from displacement and are therefore less disposed to leave their place of origin.

Households that are more established in a region and play a more important role in the community (V_i) are more popular targets for armed groups because they serve as an example to intimidate others. Moreover, by targeting community leaders, armed groups can reduce the chances of an organized community resistance. This implies the following hypotheses for the proxies used:

Years lived in the region. The longer a household has lived in the region, the more likely we would expect it to be threatened by armed groups.

Age of the household head. The impact of age is less clear. On the one hand, we might expect older household heads to play a more important role in the community and therefore to be more likely to be threatened. On the other hand, young adults may be more able to resist violent attacks and might therefore be seen as a potential threat by the armed groups.

Membership in local organizations. We would expect members of organizations to be more likely to be threatened by armed groups.

The armed actors (A_i) considered are:

Guerrilla and paramilitaries. We expect the probability of a household to be threatened to increase with the presence of these groups in the region.

Military and government security forces. On the one hand, presence of the military or other government security forces may provide some protection to the civil population against being threatened by subversive groups. On the other hand, the military might itself pose threats or increase the pressure on subversive groups.

C. Descriptive Statistics

Table 2 provides summary statistics for the exogenous variables and the incidence of direct threats for the displaced and nondisplaced population. The statistics are weighted to correct for sampling bias (see the appendix). The survey of the displaced solicited information about the presence of armed groups

in the region. The presence of right-wing paramilitary groups and left-wing guerrilla groups is reported with a greater frequency by displaced households in contrast to nondisplaced households. However, protection of government forces, like military, police, and other institutions, seems effective in deterring displacement.

Table 2 also shows the incidence of violent events suffered by the interviewed households, their friends, and their family, as well as the household's knowledge of such events in their own or neighboring communities. In general, the results are indicative of the immense exposure to violence that people in these conflict zones are exposed to. When asked about acts of violence suffered directly by household members, 78% of the displaced interviewed indicated that they or other members of their household had received death threats. As expected, the corresponding percentage is much lower (9%) for the sample of nondisplaced. Thus, the results support our earlier hypothesis. Violence plays a significant role in the displacement process. However, some households remain in their area of origin despite being quite strongly affected by the violence there because violence may not be directly targeted to them.

Violence can be less direct as well. We would expect that violent events suffered by others in the same or nearby communities would also affect households' perceptions of insecurity and own risks. Again, the survey results indicate that this indirect exposure to violence is less for the sample of nondisplaced than for the displaced. However, once more, it is quite substantial for the nondisplaced as well. For example, more than one-third of the latter had friends and one-fifth had relatives who were assassinated; almost 6% had friends who had received death threats.

The loss of property, in particular land, held at the place of origin can be another important impact of displacement. The survey results indicate that 80% of the displaced held land for agricultural purposes at their place of origin. Of these, 97% relied on the land as the principal source of household income and sustenance. Moreover, of those who held land, 58% legally owned the land and another 2% were owners without a legal title. Near 2% of the interviewed were able to sell or rent their land, while more than 94% indicated that they abandoned the land or it was taken from them. Finally, we found that 66% of the households owning land at the place of origin had received death threats, while this was the case for only 12% of the landless households.¹⁶

Displacement implies the loss of a way of life and the social network at the place of origin. We would expect this impact to be stronger the longer a person has lived there. The results presented in table 2 indicate that displaced

¹⁶ Results are based on the joint sample of displaced and nondisplaced.

households resided an average of 15 years in the origin site before displacement. Thus, for these people, displacement implies loss of the place they have considered their home for a long time and loss of the social ties they have established there. Membership to local organizations at the place of origin can be another indicator of the bondage of a household to the community they lived in for so many years. We found that 15% of the displaced households surveyed had members who were part of some organization at their place of origin. Moreover, the figures in table 2 show that displaced households are much more inclined to participate in community organizations than nondisplaced households, which may indicate that rebel groups target active members of the community with the objective of destroying social networks.

Displacement entails, as well, a descent in economic status. First, location-specific assets are usually abandoned. Second, access to social services (e.g., education and health services) is relinquished. Finally, physical and social investments are not transferable during migration. Table 2 reveals that the better-off households are less likely to displace. When compared to displaced households, nondisplaced households more often have access to social services, are better educated and, on the whole, exhibit higher economic status.

IV. Econometric Results

The survey data from the samples of displaced and nondisplaced were used to empirically assess the determinants of the displacement decision and of the probability of a household being threatened by violent actors. We used a linear specification and assumed a logistic distribution of the random error terms. To correct for choice-based sampling bias, observations were weighted, as explained in the appendix. In general, the regression results were reasonably robust to changes in the set of independent variables included. Below we first present the results on the determinants of displacement and then discuss the results for the determinants of direct threats. This is done because the latter are useful in interpreting the results of the main regression, that is, the determinants of displacement.

A. Determinants of Displacement

The results from the econometric analysis of the displacement decision are presented in table 3. Four regressions are estimated. The first regression estimates the displacement decision omitting all violence variables. The second regression includes the proxy variables for violence discussed earlier. The third regression allows for violence affecting the impact of the major economic variables by further including interaction terms. Finally, a potential limitation of our analysis may be that it assumes that the level of violence in the com-

munity and government responses are exogenous. While they are indeed likely to be exogenous to the household's displacement decision, they may depend on other community factors (e.g., community wealth and overall economic opportunities in the area). This implies that violence variables may be picking up effects of other community-level factors influencing migration through more conventional channels. To control for such possible biases from ignoring community factors and to test for the robustness of our results on the impacts of violence, the fourth regression in table 3 includes municipality variables, specifically, distance to the nearest important market and the index of basic needs.

The results support the hypothesis that security considerations play a significant role in determining displacement. When security considerations are ignored in the estimation, the standard economic variables are statistically significant, but some have the opposite sign expected in migration models (e.g., better-educated households are less likely to displace). Once security considerations are included, the influence of economic variables increases, the pseudo R^2 almost doubles and the size and significance of the constant term decreases significantly, which indicates that ignoring the violence variables causes a serious omitted variable bias. As expected, the inclusion of violence variables shows that a household is significantly more likely to opt for displacement if it has knowledge of violent acts in the close area surrounding the household (including nearby villages). The probability of displacement also increases significantly with the presence of subversive groups in the region, both paramilitary groups and guerrilla. All of these variables are significant at the 1% level in both regressions. The results suggest that the presence of both military forces and national police seem to provide some sense of security and significantly reduce the probability of displacement. Again, this effect is significant at the 1% level in both regressions.

As we will discuss next, the econometric results also support the hypothesis that security considerations are not the only determinants of the displacement decision. Through the use of interaction terms, the impact of economic incentives on the displacement decision can be differentiated for households facing high levels of indirect violence (measured as those households knowing of other incidences of violent acts in own or nearby villages) and those that do not. The results support the hypothesis that impacts of economic variables on the displacement decision differ depending on the degree of indirect violence faced by a household.

Land size has a negative coefficient for both households facing higher and lower levels of violence, but the effect is stronger for households facing lower levels of violence. There are two counteracting effects here. First, a larger

TABLE 3
ECONOMETRIC ANALYSIS OF THE PROBABILITY OF DISPLACEMENT

Variable	(1)	(2)	(3)	(4)
Indirect violence		3.7933*** (8.88)	4.3801*** (4.67)	3.0451*** (2.53)
Presence of military forces		-2.1257*** (-4.83)	-2.3218*** (-5.40)	-1.4712*** (-2.73)
Presence of police		-1.1706*** (-3.11)	-1.1122*** (-3.24)	-1.4276*** (-4.02)
Presence of paramilitary groups		3.5822*** (7.46)	1.5030*** (4.63)	3.2062*** (8.32)
Presence of guerrilla		1.4165*** (4.43)	3.4898*** (8.17)	1.6772*** (5.78)
Land ownership	1.0184*** (2.62)	.4648 (1.57)		
Land ownership × Indirect violence			.5661** (2.01)	.9914** (2.42)
Land ownership × No indirect violence			-3.2247 (-1.55)	-4.0648** (-2.34)
Land size (standardized)	-.5809*** (-4.25)	-.6311* (-1.75)		
Land size standardized × Indirect violence			-.3475 (-1.45)	-.0718 (-.19)
Land size standardized × No indirect violence			-9.0542 (-1.45)	-10.4599** (-2.16)
Access to social services	-.0294 (-.09)	-1.0943*** (-2.56)		
Access to social services × Indirect violence			-1.0400*** (-2.59)	-.4163 (-1.03)
Access to social services × No indirect violence			-3.4227*** (-6.39)	-3.3019*** (-4.5)
Highest level of education in household	-.1310*** (-3.79)	-.2230*** (-5.41)		

Highest level of education in household × Indirect violence			-.2350*** (-5.72)	-.1968*** (-4.65)
Highest level of education in household × No indirect violence			.0628 (1.13)	.0028 (.04)
Number of livestock at place of origin	1.7911*** (3.01)	2.4913*** (3.14)	1.5999*** (3.68)	1.8213 (2.34)
Access to public utilities	-1.6259** (-2.15)	-1.8817** (-2.17)	-1.4643* (-1.75)	-.9051 (-1.47)
Access to media	-.2543*** (-2.58)	-.2859*** (-2.52)	-.2662** (-2.34)	-.2896** (-2.43)
Age household head	-.0659*** (-6.26)	-.0730*** (-6.24)	-.0745*** (-6.80)	-.0696*** (-5.94)
Male household head	.0666 (.28)	-.1784 (-.71)	-.2413 (-.91)	-.5266* (-1.64)
Number of organizations	.7493*** (3.98)	1.2674*** (4.87)	1.3175*** (4.89)	1.2325*** (4.22)
Contacts at reception site	-.1673 (-.78)	.2914 (1.04)	.2443 (.85)	.0448 (.13)
Years of residence	.0110 (1.19)	.0302*** (2.99)	.0319*** (3.18)	.0281*** (2.77)
Distance to important markets (km)				.0119*** (4.69)
Index of basic needs in rural areas				.0274*** (2.64)
Constant	8.3284*** (6.99)	4.6856*** (3.80)	3.8987*** (3.48)	-2.8970 (-1.42)
Pseudo R^2	.3643	.6411	.6472	.6762
Number of observations	351	346	346	345

Note. The t-statistics are in parentheses.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

landholding tends to deter displacement as the potential loss of this important asset represents an opportunity cost to the household (direct effect). Second, our results on the determinants of direct threats, shown below, indicate that larger landholdings increase the probability for the household of receiving direct threats, which is likely to lead to an increase in the probability of displacement (indirect effect). Thus, the result that the coefficient on land size in table 3 is less negative for households faced with higher levels of violence is consistent with the idea that for these households economic considerations play a less important role relative to security considerations as compared to households facing lower violence levels. Similarly, it is found that land ownership increases the probability of displacement for households confronting high levels of violence, whereas land ownership deters displacement for households faced with lower levels of violence, although the latter effect is only significant at the 12% level.

Other aspects of the standard of living appear to have an impact on the displacement decision as well. As expected, we found that those households with access to social services (education and health) and connection to public utilities at their place of origin are significantly less likely to opt for displacement. The coefficient on social services is, however, three times larger for households not confronting violence. The difference in coefficients between households facing higher and lower levels of violence is significant at the 1% level.

The number of livestock owned, however, has a significantly positive effect on the probability of displacement. As discussed in Section III.B, this may be due to the fact that livestock is a form of savings that can be transformed into cash relatively easily (much more easily than land). Thus, livestock ownership may provide financial resources that help to cover the costs of displacement and provide a basis for a new start in the receptor location.

The age of the household head has a significant negative effect on displacement. This is consistent with the idea that younger household heads have longer planning horizons and are less attached to the region of origin. Households headed by women are more likely to displace, although the effect is not significant. The variable "contacts at reception site" showed a positive sign but did not have a significant effect in the regressions. This is, however, not surprising because personal contacts tend to reduce the immediate costs of migration but also may reduce expectations on the standard of living after displacement. The results indicate that these effects partly offset each other, although the positive coefficient suggests that the former effect dominates.

The results on the variables reflecting access to information appear to support

the hypothesis that improved information actually leads to a lower probability of displacement because it lowers overly optimistic expectations regarding the conditions encountered by the displaced in the receptor cities. This result stands in contrast to the migration literature, where improved information is usually associated with a higher propensity to migrate. More specifically, our results indicate a significant negative impact of a household's access to public media on the probability of displacement. This may be due to the frequent reports in the Colombian media on the precarious situation of the displaced. Moreover, the household head's level of education has a significant negative impact on displacement for households facing high levels of violence. This could reflect the idea that more educated persons have more realistic expectations regarding the situation of the displaced. This information effect appears to dominate the "standard of living effect," that is, the idea that higher education levels tend to raise the expected income in the receptor city and thereby increase the likelihood of displacement. By contrast, the standard of living effect seems to be stronger for households facing lower levels of violence. For these households, education has a positive but not significant effect. Thus, once again, our results support the hypothesis that the impact of economic considerations is stronger where violence is less pronounced. (The difference in coefficients is significant at the 1% level.)

The econometric results do not support the hypothesis that households with stronger ties to the region of origin have a lower propensity for displacement, except for the result on age described earlier. It is found that households with more years of residence and those whose members belong to a larger number of organizations are more likely to displace. One possible explanation is that long-term residents and members of organizations may have extensive social networks in the place of origin that can assist them during displacement by looking after their properties. This support system may prevail for a period, and therefore, displacement does not necessarily mean losing the ties in the region of origin in this sense but rather may decrease the costs of displacement.

Finally, community factors also play a role in the displacement decision. As expected, lack of economic opportunities in the communities of origin appears to encourage the decision to displace. Households living previously in communities far from economic markets or with high indices of basic needs are more likely to displace. Moreover, incorporating community factors decreases the influence of indirect violence and the presence of a military force somewhat. Yet security considerations continue to be highly significant determinants of the displacement decision.

TABLE 4
ECONOMETRIC ANALYSIS OF THE PROBABILITY OF DIRECT THREATS

Variable	Coefficient	t-Statistic
Indirect violence	.2434	.25
Presence of military forces	1.8613***	3.13
Presence of police	-1.0732**	-2.02
Presence of paramilitary groups	.1749	.23
Presence of guerrilla	1.1995*	1.74
Land ownership	-.0684	-.12
Land size standardized	.8465**	2.41
Access to social services	-.6969	-1.34
Highest level of education in household	.0888	1.13
Number of livestock at place of origin	.5707	.76
Access to public utilities	-1.2921**	-2.45
Access to media	.6213***	3.05
Age household head	.0080	.43
Male household head	-1.1262**	-2.34
Number of organizations	-.3269	-1.61
Contacts at reception site	-.4250	-.89
Years of residence	.0052	.32
Constant	4.6865	1.23
Pseudo R^2	.5705	
Number of observations	346	

Note. The dependent variable is direct threats (equals 1 if any household member suffered death threats, 0 otherwise).

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

B. Determinants of Direct Threats

The results from the econometric estimation of the probability of being a victim of direct threats are presented in table 4. As expected, the presence of guerrilla and paramilitary groups increases the probability of direct threats, but the effect is only statistically significant for paramilitary groups.¹⁷ It is interesting that the presence of military also has a significant and positive effect on the probability of being threatened. This is consistent with some studies by nongovernmental organizations (NGOs) and other research on violence. By contrast, the presence of police forces in the region significantly decreases the likelihood of threats. This seems to indicate that nonmilitary government presence does deter subversive groups from posing direct threats to specific households in the community to some extent. It is important to conduct further research to assess whether our result can be generalized. Despite

¹⁷ Two important caveats are in order about this result. First, after the survey was conducted, the dynamics of the war changed dramatically. In 2003, guerrilla groups displaced civil population in similar proportions as paramilitary groups, whereas, in 2000, paramilitary groups were the main actors responsible for displacement. Second, while guerrilla groups displaced population using armed confrontations, paramilitary groups used direct threats. Therefore, the greater significance of this coefficient is not surprising.

the effect of military presence in increasing the likelihood of direct threats, the results from the displacement regression discussed earlier indicated that military presence deters displacement. A possible interpretation is that the permanent presence of two or more armed forces contesting a territory increases the pressure on the population. However, if displacement is imminent and military forces arrive to protect the population, their presence may deter displacement.

The results also support our hypothesis that households owning location-specific assets are more likely to be threatened by violent actors. Specifically, as was mentioned earlier, landholding size has a significant positive effect on the probability of being threatened. For other dimensions of economic status the effect is less clear. In particular, access to education and health services as well as public utilities shows a negative impact on the probability of being threatened, although the effect is only significant in the case of the latter.

The results also indicate that households headed by women are significantly more likely to be threatened than those headed by men. This supports the statements often found in the literature that women are the prime victims of violence. This interpretation is to be taken with a caveat, however, as other studies also suggest that households often become headed by women as a consequence of displacement and violence when male household heads get assassinated or leave their families after displacement (Ibáñez and Querubin 2004). Moreover, the insignificant effect of gender in the displacement regressions indicates that the increased probability for women of being threatened is partially counteracted by other effects, that is, the fact that women may have lower expected earnings in the receptor cities. Unfortunately, the reduced-form estimation of the displacement regression does not permit a precise distinction of these potential effects.

The hypothesis that households that are more established in the region and that play a more important role in the community are more likely to be threatened directly is only partly supported by the results. The years a household has lived in the region of origin has a positive effect on threats, although the effect is not significant. Counter to our expectations, the number of organizations a household belongs to has a negative impact on threats, and the effect is significant at the 10% level. This result is somewhat puzzling as other studies suggest that community leaders are more likely to be targeted by subversive groups (Henao 1998; Lozano and Osorio 1999). The result may be partly due to the fact that numbers of organizations is an imperfect proxy for the social status of the household in the community. Another possible explanation is that organizations may be helpful in protecting its members to some degree, reducing the probability of threats.

V. Conclusions and Policy Implications

Obviously, any real solution of the displacement problem requires the end of violent conflict. However, as long as peace remains unlikely in Colombia, as well as in many other cases, we need to consider other options to alleviate the problem. An improved understanding of the determinants and effects of displacement is an important prerequisite for the definition of appropriate policies of prevention, assistance, and relocation.

This article contributes to such an understanding through the development of a conceptual framework for the empirical analysis of the determinants of displacement decisions at the household level. While similarities to conventional migration theories and their hypotheses exist, our discussion shows that there are also important differences implied by the presence of violent conflicts. Some considerations, such as risk aversion and lack of information, may enhance conventional migration but hamper displacement. Moreover, factors such as landholdings and social capital at the place of origin may not only reduce displacement in the conventional sense of increasing opportunity costs but also increase displacement indirectly by increasing the household's likelihood of becoming the victim of direct threats.

Focusing on the case of Colombia, the conceptual framework has permitted us to provide a more rigorous assessment of hypotheses on the specific factors involved in displacement decisions as well as the determinants of becoming a victim of direct threats by armed groups. Our analysis has confirmed the very important role of violence and perceptions of insecurity in motivating displacement. The presence of government security forces may help to reduce insecurity in areas of high violence. Given that direct threats are probably an extremely important trigger of displacement, government protection should also concentrate on those groups that are most at risk of being threatened. The econometric results show that households with larger landholdings and those with female household heads carry a higher risk of being the target of such threats.¹⁸

Security considerations, however, are not the only factors underlying the displacement decision. People decide to stay in their areas of origin despite being affected by violence. A significant percentage of the displaced appear to have reflected on their options and the expected impacts of displacement. Considerations regarding the cost of abandoning important assets, particularly land for agricultural production, appear to affect displacement decisions. Sim-

¹⁸ It should be stressed that our analysis was focused on lower-income households with fairly small landholdings. Large landowners are potentially more able to provide for their own protection.

ilarly, households living without access to public utilities, health services, or child education are more likely to opt for displacement.

Our results indicate that economic incentives play a role for households' displacement decisions even in a context of violence but that the impact of economic incentives is less strong where violence levels are high. Thus, while violence tends to increase the urbanization trend observed in Colombia directly, it reduces the responsiveness of urbanization to economic incentives. One implication of this result is that policies aimed at reducing the imbalance between rural and urban economic opportunities may be less effective in reducing urbanization in a context of violence.

The results also point to the important role of information in displacement decisions. It appears that expectations regarding the conditions in receptor cities are sometimes overly optimistic. Improved information on the problems encountered after displacement reduces households' propensity to opt for displacement. Therefore, an important policy recommendation is to improve the flow of such information. Of course, policies should not stop there but rather need to also provide alternatives to the households who feel threatened by violence. In addition to the security considerations discussed above, one option might be to build up a more decentralized support network closer to the expulsory regions. This could also help to reduce the immense pressure on some of the current receptor cities. Current provisions within Colombia's Law 387 that promote the Municipal and Departmental Committee of Attention as the organism in charge within the National System of Integrated Attention of Violently Displaced People are a move in this direction, but, in practice, the majority of the displaced do not yet receive support from these committees.

Households that opt for displacement for preventive reasons are more likely than those reacting to sudden threats to consider the potential advantages and disadvantages of alternative options, including the choice of receptor location. They may also be able to avoid some of the losses implied in a sudden departure. Moreover, policies of return may be more viable for these households than for those who did not have an opportunity to organize their departure and who experienced the traumas of extreme violence. Therefore, improved information predicting future "hotspots" of violence and assessing alternative destinations would be desirable. The Displacement Information System supported by the United Nations, whose main objective is to issue alerts on possible displacement incidents, monitor the number of displaced persons, and work with households in the process of displacement, appears to be a move in the right direction in this regard.

One of the objectives proposed by Law 387 is to create and apply mechanisms that provide legal and judicial assistance to the displaced population in order

to guarantee the investigation of events, the restitution of violated rights, and the defense of the affected goods (Law 387 from 1997, chap. 2, sec. 1). Although the national government already created mechanisms to protect the land of displaced population by issuing Decree 2007 from 2001, it is important to generate more aggressive policies to establish the actual state of the abandoned land properties, to find out who is using them at the moment, and to generate a process of recognition of property rights.

Finally, several interesting and interrelated topics for future research emerge. First, one might consider representing the displacement process as dynamic in the following sense. When violent actors first move into a region, some of the civil population gets taken by surprise, is attacked, and then will respond “reactively.” Other members of the community observe the violent acts and may then respond “preventively.” Second, an interesting theoretical consideration is that, in the long run, we might expect households to consider the risk of displacement in their choice of assets. If violence prevails long enough in the region, households may consider investing more in mobile assets (e.g., human capital) and less in location-specific assets (e.g., land and housing). Testing for this potential long-run endogeneity would be an interesting subject for research. Similarly, one could also think of modeling the displacement process as a game between violent groups and the local community and examine the conditions that allow communities to resist displacement.

Appendix

Computation of Weights

As stated earlier, we followed the approach suggested in Manski and Lerman (1977) and uses a weighted exogenous sampling maximum likelihood estimator in our econometric analysis. To construct the weights for the displaced sample, we calculated the fraction of the displaced population selecting each municipality where the survey was conducted (Q_i^d) and the analogous fraction for the choice-based sample (H_i^d).¹⁹ Average statistics on these variables are presented in table A1. The weight applied to contribution i of the log-likelihood for the displaced sample is

$$w_i^d = Q_i^d / H_i^d.$$

Similarly, the weight for the sample of nondisplaced households is

$$w_i^{nd} = Q_i^{nd} / H_i^{nd},$$

where Q_i^{nd} represents the fraction of the population that stayed in the municipality

¹⁹ These figures are available at <http://www.red.gov.co>.

where the survey was conducted and H_i^{nd} is the analogous fraction for the choice-based sample. By weighting each contribution of the log-likelihood by these weights, unbiased parameter estimates are obtained.

TABLE A1
STATISTICS USED FOR COMPUTATION OF MANSKY WEIGHTS

Variable	Mean
Number of displaced persons between 1995 and 1999	1,268
Population in 1999	64,986
Percentage of displaced population	1
Percentage of nondisplaced population	99

Source. Authors' calculations based on municipal data collected at Centro de Estudios sobre Desarrollo Económico (2005).

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